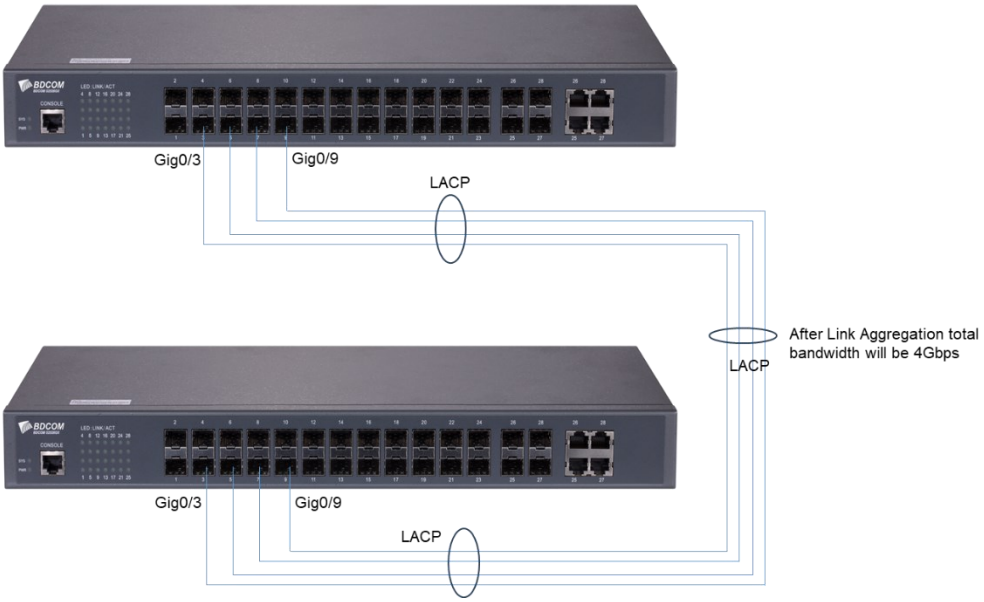
Link Aggregation Control Protocol (LACP)

This document describes how to configure combine multiple physical network links into a single logical link to increase bandwidth and provide redundancy in BDOM switches. It operates according to IEEE 802.3ad (also known as IEEE 802.1AX).

There are Two Configuration Modes:

1. LACP (Active/Passive)
2. Static

	LACP (Active/Passive)	Static
Negotiation and Management	Uses LACP to negotiate and manage the aggregation dynamically. Active mode initiates and responds to LACP packets, while passive mode only responds.	No dynamic negotiation; requires manual configuration. There is no dynamic management. All changes must be handled manually by the network administrator.
Fault Tolerance	Provides better fault tolerance and automatic recovery. If one link in the aggregation fails, LACP can dynamically adjust and reconfigure the remaining links.	Does not provide automatic failover. If a link fails, the network administrator must manually reconfigure the link aggregation. Lacks automatic adjustment and recovery. If a link fails, manual reconfiguration might be required.
Configuration Complexity	More complex to set up due to the dynamic nature but offers more flexibility and automation.	Simpler to configure but less flexible and requires manual intervention for changes or fault recovery.
Compatibility	Requires support for LACP on both ends of the link.	Can be used with devices that do not support LACP but requires identical manual configurations on both ends.



Topology: Link Aggregation Control Protocol (LACP)

Configuration of Switch-1:

- Switch1#config
- Switch1_config#interface port-aggregator 1
- Switch1_config_p1#switchport mode trunk
- Switch1_config_p1#switchport trunk vlan-allowed 100,200,300

```
COM3 - PuTTY
Switch1#
Switch1#
Switch1#
Switch1#
Switch1#
Switch1#config
Switch1_config#interface port-aggregator 1
Switch1_config_p1#switchport mode trunk
Switch1_config_p1#switchport trunk vl
vlan-allowed  vlan-untagged
Switch1_config_p1#switchport trunk vlan-allowed 100,200,300
Switch1_config_p1#
```

- Switch1#config
- Switch1_config#interface gigaEthernet 0/1
- Switch1_config_g0/1#aggregator-group 1 mode lacp
- Switch1_config_g0/1#switchport mode trunk
- Switch1_config_g0/1#switchport trunk vlan-allowed 100,200,300

- Switch1#config
- Switch1_config#interface gigaEthernet 0/3
- Switch1_config_g0/3#aggregator-group 1 mode lacp
- Switch1_config_g0/3#switchport mode trunk
- Switch1_config_g0/3#switchport trunk vlan-allowed 100,200,300

- Switch1#config
- Switch1_config#interface gigaEthernet 0/5
- Switch1_config_g0/5#aggregator-group 1 mode lacp
- Switch1_config_g0/5#switchport mode trunk
- Switch1_config_g0/5#switchport trunk vlan-allowed 100,200,300

```
COM3 - PuTTY
Switch1_config#interface gigaEthernet 0/1
Switch1_config_g0/1#aggregator-group 1 mode lacp
Switch1_config_g0/1#switchport mode trunk
Switch1_config_g0/1#switchport trunk vlan-allowed 100,200,300
Switch1_config_g0/1#
Switch1_config_g0/1#exit
Switch1_config#interface gigaEthernet 0/3
Switch1_config_g0/3#aggregator-group 1 mode lacp
Switch1_config_g0/3#switchport mode trunk
Switch1_config_g0/3#switchport trunk vlan-allowed 100,200,300
Switch1_config_g0/3#exit
Switch1_config#
Switch1_config#interface gigaEthernet 0/5
Switch1_config_g0/5#aggregator-group 1 mode lacp
Switch1_config_g0/5#switchport mode trunk
Switch1_config_g0/5#switchport trunk vlan-allowed 100,200,300
Switch1_config_g0/5#exit
Switch1_config#wr
Saving current configuration...
OK!
Switch1_config# 1  1 00:10:51 /startup-config is wrote, TID:84e7c830
```

Configuration of Switch-2:

- Switch2#config
- Switch2_config#interface port-aggregator 1
- Switch2_config_p1#switchport mode trunk
- Switch2_config_p1#switchport trunk vlan-allowed 100,200,300

```
COM3 - PuTTY
Switch2>
Switch2>enable
Switch2# 1  1 01:14:34 User admin enter privilege mode from console 0, level = 1
5

Switch2#config
Switch2_config#interface port-aggregator 1
Switch2_config_p1#switchport mode trunk
Switch2_config_p1#switchport trunk vlan-allowed 100,200,300
Switch2_config_p1#exit
Switch2_config#
```

- Switch2#config
- Switch2_config#interface gigaEthernet 0/2
- Switch2_config_g0/2#aggregator-group 1 mode lacp
- Switch2_config_g0/2#switchport mode trunk
- Switch2_config_g0/2#switchport trunk vlan-allowed 100,200,300

- Switch2#config
- Switch2_config#interface gigaEthernet 0/4
- Switch2_config_g0/4#aggregator-group 1 mode lacp
- Switch2_config_g0/4#switchport mode trunk
- Switch2_config_g0/4#switchport trunk vlan-allowed 100,200,300

- Switch2#config
- Switch2_config#interface gigaEthernet 0/6
- Switch2_config_g0/6#aggregator-group 1 mode lacp
- Switch2_config_g0/6#switchport mode trunk
- Switch2_config_g0/6#switchport trunk vlan-allowed 100,200,300

```
COM3 - PuTTY
Switch2_config_g0/2#aggregator-group 1 mode lacp
Switch2_config_g0/2#switchport mode trunk
Switch2_config_g0/2#switchport trunk vlan-allowed 100,200,300
Switch2_config_g0/2#exit
Switch2_config#
Switch2_config#interface gigaEthernet 0/4
Switch2_config_g0/4#aggregator-group 1 mode lacp
Switch2_config_g0/4#switchport mode trunk
Switch2_config_g0/4#switchport trunk vlan-allowed 100,200,300
Switch2_config_g0/4#
Switch2_config_g0/4#exit
Switch2_config#interface gigaEthernet 0/6
Switch2_config_g0/6#aggregator-group 1 mode lacp
Switch2_config_g0/6#switchport mode trunk
Switch2_config_g0/6#switchport trunk vlan-allowed 100,200,300
Switch2_config_g0/6#
Switch2_config_g0/6#exit
Switch2_config#wr
Saving current configuration...
OK!
Switch2_config# 1  1 00:14:41 /startup-config is wrote, TID:84e7c830
```

Configuring Traffic Load Balance After Port Aggregation:

We need to configure a proper load-balancing mode based on traffic characteristics. When a parameter of traffic changes frequently, we can set the load balancing mode based on this parameter to ensure that the traffic load is balanced evenly. We may use the load balancing mode based on following parameters

src-mac -	Source MAC Address
dst-mac-	Destination MAC Address
both-mac-	Source and Destination MAC Address
src-ip-	Source IP Address
dst-ip-	Destination IP Address
both-ip-	Source and Destination IP Address
src-port-	Source Port
l4-sport-	L4 Source Port
l4-dport-	L4 Destination Port
both-l4port-	L4 Source and Destination Port

```
Switch2_config#
Switch2_config#aggregator-group load-balance  ?
  src-mac       -- Src Mac Addr
  dst-mac       -- Dst Mac Addr
  both-mac      -- Src and Dst Mac Addr
  src-ip        -- Src Ip Addr
  dst-ip        -- Dst Ip Addr
  both-ip       -- Src and Dst Ip Addr
  src-port      -- Src port
  l4-sport      -- L4 Src port
  l4-dport      -- L4 Dst port
  both-l4port   -- L4 Src and Dst port
Switch2_config#
Switch2_config#
Switch2_config#
Switch2_config#
Switch2_config#
Switch2_config#
Switch2_config#
Switch2_config#
Switch2_config#
Switch2_config#
Switch2_config#
Switch2_config#aggregator-group load-balance both-l4port
Switch2_config#
Switch2_config#
```

Display the specific information of aggregator-group:

#show aggregator-group [id] {detail | brief | summary}

- show aggregator-group 1 brief
- show aggregator-group 1 detail
- show aggregator-group 1 summary

Note: We can use any VLAN instead of 100, 200, 300. If doesn't require then no need to use VLAN. For BDCOM switches it can be created 8 to 16 groups (depending on the models) and under each group can be created maximum 8 links to aggregate (depending on the models)

The aggregator group must be the same if both switches have multiple port aggregators. For example: Switch1- P4 & Switch2- P4 to ensure the configuration identical parameters support LACP Modes.

Sometimes there could be an error like below –

Partner port aggregator timer is inconsistent, change the port(tg0/3) timer!!!

In this case we may set the timer to ensure both sides have to same timer

```
Switch#config
Switch_config#interface port-aggregator 1
Switch_config_p1#agg-period ?
  <1-30>     -- Aggregation negotiation period timer, default: 30S
```

The END